

Problem-Solution Fit

Date	04 July 2026
Team ID	
Project Name	Rising Waters – ML-Based Flood Prediction System
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Meteorology & disaster-management officials; district disaster-relief coordinators; government agricultural/water-resource analysts; farmers and residents in flood-prone river-basin districts of India.	6. CUSTOMER LIMITATIONS (e.g. budget, devices) [CL] Limited budget for specialist forecasting software; rural areas with patchy internet/low-end smartphones; little technical/statistical training to interpret raw meteorological data.	5. AVAILABLE SOLUTIONS – PROS & CONS [AS] Govt. weather bulletins: authoritative but broad/delayed, not location-specific. News/TV flood alerts: reach many, but reactive not predictive. Manual statistical analysis: accurate but slow, needs a data-science expert each time.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] No quick, localized way to estimate flood risk from current rainfall/weather readings; warnings often arrive after flooding starts. Recurs every monsoon season (roughly June–September) and after any unusually heavy rainfall spell.	9. PROBLEM ROOT / CAUSE [RC] Flood risk depends on many interacting seasonal-rainfall and atmospheric variables; without a trained model to combine them, assessment stays manual, slow, and inconsistent across regions/officials.	7. BEHAVIOR + ITS INTENSITY [BE] Frequently checking rainfall totals, humidity and cloud-cover reports during the monsoon; intensity spikes sharply whenever seasonal rainfall approaches or exceeds historical flood-triggering levels.

3. TRIGGERS TO ACT [TR] Onset of the monsoon; a sudden jump in daily/seasonal rainfall totals; official heavy-rain advisories; rising humidity and cloud cover consistent with past flood events.	10. YOUR SOLUTION [SL] “Rising Waters”: a Flask web app backed by a trained ML classifier (XGBoost, 95.65% test accuracy) that takes temperature, humidity, cloud cover and seasonal rainfall figures and instantly returns a flood-risk prediction with probability, so at-risk regions can act before flooding occurs.	8. CHANNELS OF BEHAVIOR [CH] Online – weather-department websites/apps, satellite rainfall dashboards, news alerts, social media flood warnings.
4. EMOTIONS BEFORE / AFTER [EM] Before: anxiety and uncertainty over whether current conditions will lead to a flood. After (using the tool): reassurance and confidence from a clear, data-backed probability that supports faster, better-informed decisions.		Offline – radio bulletins, district disaster-management announcements, community loudspeaker/mobile alerts in rural areas.